PhD Lecture in Behavioral Asset Pricing: Preferences & Beliefs (II)

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Psychology of Beliefs

Psychology of Beliefs: Representativeness

- Consider questions such as:
  - What is the probability that object A comes from class B?
  - What is the probability that event A was generated from process B?
- Kahneman and Tversky (1974) argue that people often answer with the "representativeness heuristic"
  - Evaluate the probability by the extent to which A is representative of B
  - i.e. the degree to which A reflects the essential characteristics of B
- This is helpful in general but can lead to serious biases

Representativeness, ctd.

- **Base-rate neglect**
- Consider the following description of Steve. Do you think Steve is more likely to be a librarian or a lawyer?
  - Steve is very shy and withdrawn, invariably helpful, but with little interest in people or in the world of reality.
  - A meek and tidy soul, he has a need for order and structure, and a passion for detail.
- People judge Steve to be more likely a librarian than a lawyer
- But there are many more lawyers than librarians in the U.S.
- The question actually asks about the LHS, but people answer using only the likelihood ratio (the first term on the RHS); this is the root cause of base-rate neglect.

$$\frac{p(\text{library} \mid \text{data})}{p(\text{lawyer} \mid \text{data})} = \frac{p(\text{data} \mid \text{library})}{p(\text{data} \mid \text{lawyer})} \frac{p(\text{library})}{p(\text{lawyer})}.$$

Representativeness, ctd.

- **Sample size neglect**
- When we do not know the model generating the data, we may infer it too quickly based on too little data
- This can lead to over-extrapolation of recent trends
- E.g., the hot hand fallacy
  - Seeing a basketball player make a few shots in a row leads to the belief that he is "hot"
  - But empirical evidence suggests that the hot hand is largely a myth (Gilovich, Vallone, and Tversky, 1985; but see Miller and Sanjurjo, 2018)

Psychology of Beliefs: Overconfidence

- There are at least two distinct types of overconfidence:
- **Overprecision**
  - People are too confident in the accuracy of their forecasts
  - 98% confidence intervals include the true quantity around 60% of the time (Alpert and Raiffa, 1982)
- **Overplacement**
  - People think they are better than others
  - 90% of drivers think they are above-average drivers (Svenson, 1981)

Overprecision: Survey Evidence

- Ben-David, Graham, and Harvey (2013)
- CFOs' expected stock market return distribution is too narrow
- Realized market returns are within the executives' 80% confidence intervals only 36% of the time

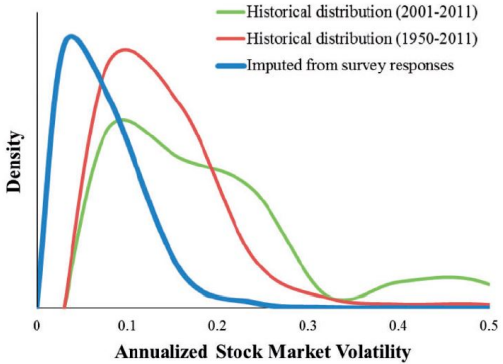


FIGURE IIA

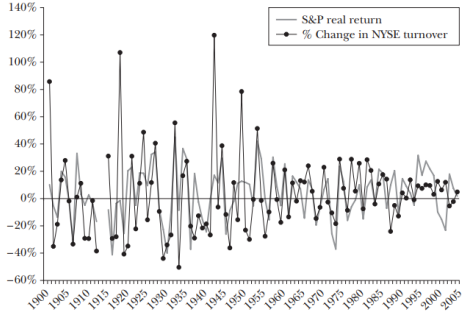
One-Year Imputed Individual S&P 500 Volatility Versus Historical Distribution

The kernel density of the one-year S&P 500 return volatility for the period of 1950–2000 (or 2001–2011), and the kernel density of the imputed one-year S&P 500 individual volatility derived from CFOs' predictions.

Overconfidence: Excessive Trading

- **Excessive trading**
  - Under efficient markets, there is no trading other than for liquidity needs and portfolio rebalancing
  - Speculative trading reflects differences of opinion and some amount of overconfidence
- Trading leads to lower returns
  - The more you trade, the lower your return (Odean, 1999)

Figure 3
Market-Wide Stock Returns and Changes in Turnover, 1901–2005



Hong and Stein (2007)

Overconfidence: Excessive Trading

- **Boys will be boys**
- Barber and Odean (2001) show that men trade more frequently than women, but earn lower returns
- Consistent with psychological studies showing that men are more overconfident than women (Lundeberg, Fox, and Punćcohař, 1994)

Dependent variable	Mean monthly turnover (%)	Own-benchmark abnormal net return	Portfolio volatility
Intercept	6.269***	−0.321***	11.466***
		(−11.47)	(58.85)
Single	0.483***	0.002	0.320***
	(4.24)	(0.14)	(3.40)
Woman	−1.461***	0.058***	−0.689***
	(−12.76)	(4.27)	(−7.27)
Single × woman	−0.733***	0.027	−0.439**
	(−3.38)	(1.08)	(−2.45)



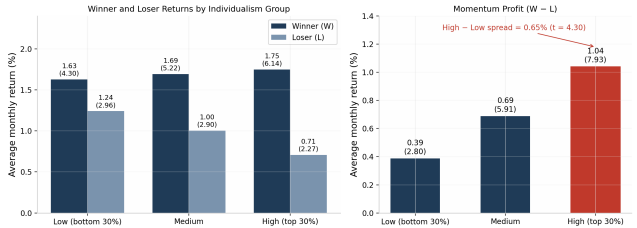
Overconfidence: Momentum

- Daniel, Hirshleifer, and Subrahmanyam (1998)
- Momentum is driven by investor overconfidence and self-attribution bias
  - Investors are overconfident in the private information they gather
  - Investors also suffer from self-attribution bias: they credit themselves for successes but blame external factors for failures
- Positive returns reinforce overconfidence, leading to more aggressive trading and price momentum
- Eventually, prices correct when information is revealed

Overconfidence: Momentum, ctd.

- Chui, Titman, and Wei (JF 2010) test the idea that momentum is driven by investor overconfidence and self-attribution bias
- They use an individualism index developed by Hofstede (2001) to proxy for investor overconfidence
- They find that momentum is stronger in countries with higher individualism scores

Table IV, Panel A — Momentum Profits by Individualism (Country-Average Portfolios, Feb 1984 – Jun 2003)
Chui, Titman & Wei (2010), Journal of Finance



Psychology of Beliefs: Over-extrapolation

- The estimate of the future value of a quantity is an increasing function of recent past values of that quantity
- One of the most important and widely applied ideas in behavioral finance
- The source of over-extrapolation: the representativeness heuristic
- Application to finance:
  - Return extrapolation: predict future returns based on past returns
  - Fundamentals extrapolation: predict future earnings/cash flows based on past growth

Over-extrapolation: A Simple Model

- **An economy with $T+1$ periods, $t = 0, 1, \ldots, T$, and two assets**
  - A riskless asset with a constant return of zero
  - A risky asset that is a claim to a single final cash flow $\tilde{D}_T = D_0 + \tilde{\varepsilon}_1 + \ldots + \tilde{\varepsilon}_T$, with $\tilde{\varepsilon}_t \sim N(0, \sigma_\varepsilon^2)$ i.i.d.
- **Some investors have extrapolative beliefs about price changes:**

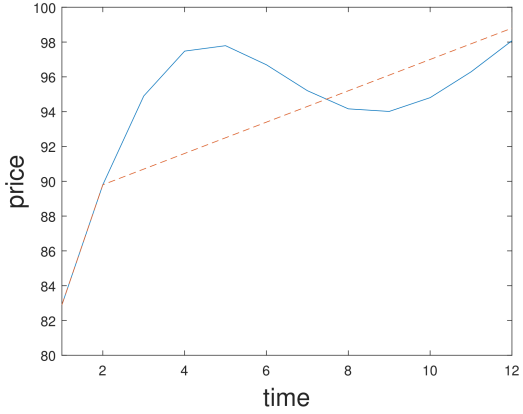
$$\mathbb{E}_t^e(P_{t+1} - P_t) = X_t \equiv (1-\theta)\sum_{k=1}^{t-1} \theta^{k-1}(P_{t-k} - P_{t-k-1}) + \theta^{t-1} X_1$$

- **Other investors — "fundamental traders" — buy when prices are low relative to expected cash flows, and sell when they are high**
- **Equilibrium prices (Barberis, 2018):**

$$P_t = D_t + \frac{\hat{\mu}}{\hat{\mu}_f} X_t - \gamma \sigma_\varepsilon^2 Q \left(T - t - 1 + \frac{1}{\hat{\mu}_f} \right)$$

Over-extrapolation: Price Dynamics

- **Price response to a positive cash-flow shock at time 2:**
  - Under-reaction at first, then continued drift up, then over-reaction and a peak, then a reversal
  - The various asset pricing applications of return extrapolation can all be seen in this figure



Over-extrapolation: Increasingly Sophisticated Models

- **Aggregate stock market:**
  - Barberis, Greenwood, Jin, and Shleifer (2015); Adam, Marcet, and Beutel (2017); Jin and Sui (2022)
- **Cross-section of stocks:**
  - Hong and Stein (1999); Barberis and Shleifer (2003); Da, Huang, and Jin (2021)
- **Bubbles:**
  - Barberis, Jin, Greenwood, and Shleifer (2018); Bastianello and Fontanier (2024)
- **Real estate market:**
  - Glaeser and Nathanson (2017)

Over-extrapolation: Open Questions

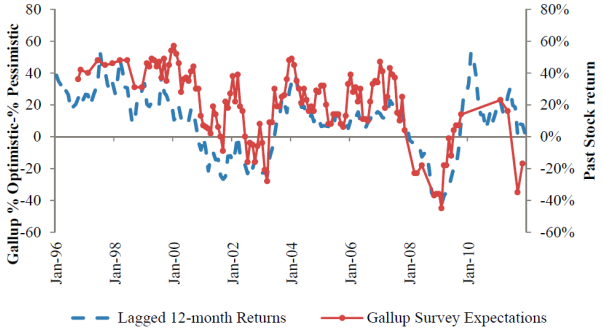
- **What is the source of return extrapolation?**
  - And can knowledge of the source deepen our understanding of the empirical facts?
- **What value of θ is consistent with observed prices, and is this value justified?**
- **Does θ vary over time, and if so, why?**
  - Cassella and Gulen (2018)
- **Are return extrapolation models consistent with the facts about investor portfolios?**
  - Who are the investors who are actually trading according to extrapolative beliefs?
  - Giglio, Maggiori, Stroebel, and Utkus (2021) find that investors' holdings are relatively insensitive to their beliefs
  - So can shifts in beliefs affect holdings enough to move prices?
  - Is this investor behavior really about beliefs — or a different, non-belief mechanism, e.g. observational learning?

Over-extrapolation: Survey Evidence

- **Excess volatility and predictability in the aggregate stock market**
- Extrapolators buy assets whose prices have risen because they expect them to keep rising
- This leads to overvaluation and return predictability

The Role of Past Stock Market Returns in Explaining Survey Expectations

The dashed line denotes the 12-month rolling nominal return on the CRSP VW stock index. The solid line marked with circles denotes expectations from the Gallup survey (% optimistic - %pessimistic).



Greenwood and Shleifer (2014)

Over-extrapolation: Survey Evidence, ctd.

- Investors act on their over-extrapolative beliefs with flows into (out of) stock markets

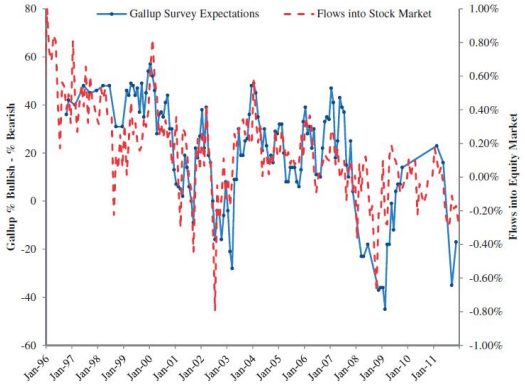


Figure 4
Comparing the Gallup survey with flows into equity mutual funds
The solid line denotes the percentage of investors who are bullish in the Gallup survey (left axis). The dashed line (right axis) is flows into mutual funds as a percentage of equity market capitalization, as reported by the Investment Company Institute.

Greenwood and Shleifer (2014)

Over-extrapolation of Past Returns: Applications

- Using survey data on expectations of stock returns, Casella and Gulen (RFS 2018) estimate the degree of extrapolative thinking in investors' beliefs (DOX) about market returns
- DOX $(1-\lambda)$ measures the weight investors place on past returns

$$Exp_t = a + b\sum_{i=0}^{\infty} w_i\, R_{t-(i+1)\Delta t,\, t-i\Delta t},$$

$$w_i = \frac{\lambda^i}{\sum_{k=0}^{\infty}\lambda^k}, \quad 0 \le \lambda < 1,$$

- A lower λ implies that investors place higher weight on more recent observations
- Earlier observations contribute less to an extrapolator's expectations

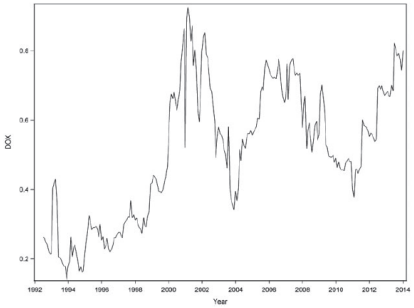
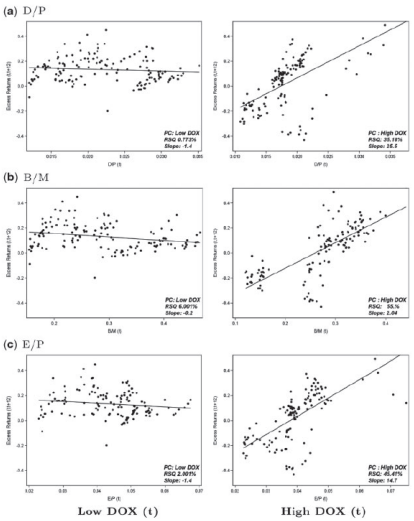


Figure 2
Time-varying DOX
The figure plots the recursively estimated DOX time series for the principal component (PC) of the II and AA surveys, in the sample period 1992:06–2013:12. (Details on estimation are in Section 2.)

Over-extrapolation of Past Returns: Applications, ctd.

- They find the predictability of stock returns by a price-scaled variable (e.g., D/P, B/M, or E/P) is contingent on the DOX
- When DOX is high, investors more quickly "forget" the positive price changes that drove up valuations, and their beliefs mean-revert faster
- As a result, high valuations are more strongly associated with low future returns when DOX is high



Extrapolative Beliefs in the Cross-Section: Setup and Data

Question. How do investors form expectations about *individual stock* returns — and are those expectations rational?

Data: Forcerank crowdsourcing platform (Feb 2016–Dec 2017):

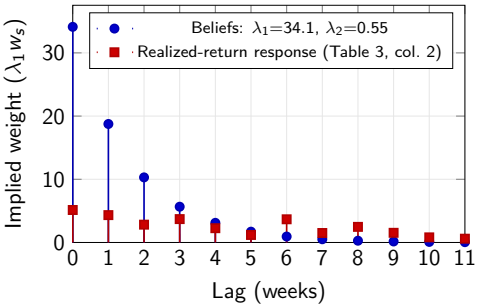
- Rank 10 same-industry stocks by expected return over the *next week*; blind setting rules out herding.
- 1,396 contests, 293 tickers, 12,798 rankings, 1,045 users (72 financial professionals).
- Consensus score = average ranking ⇒ investor expectation proxy.

Key object. Exponential-decay beliefs:

$$\mathbb{E}_i^e[R_{i,t+1}] = \lambda_1 \sum_{s=0}^{11} w_s R_{i,t-s}, \qquad w_s = \frac{\lambda_2^s}{\sum_j \lambda_2^j}, \quad 0 \le \lambda_2 < 1$$

λ_1: level of extrapolation; λ_2: recency (lower ⇒ more weight on recent returns).

Estimated belief weights on past weekly returns



Illustrative reconstruction from reported estimates; weights decay sharply with recency.

Extrapolative Beliefs: Asymmetry and Heterogeneity

Negative returns weigh more and decay slower.

- Overall: return 4 weeks back $\approx 9\%$ as important as last week's.
- After *negative* returns: 4 weeks back $\approx 45\%$ as important — investors dwell on losses.
- Same asymmetry at the market level: λ_2 higher in down weeks (cf. Cassella–Gulen 2018, bear vs. bull markets).

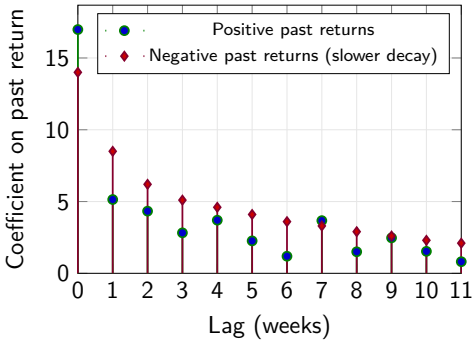
Salience channel. Dispersed contest cross-sections, heavy news coverage $\Rightarrow$ stronger ($\lambda_1 \uparrow$) and longer-lived ($\lambda_2 \uparrow$) extrapolation.

Who extrapolates more?

Group	λ_1 (level)	λ_2 (recency)
Nonprofessionals	High	Low
Financial professionals	*significantly lower*	*higher*
Large / low-B/M stocks	Higher	Higher

Professionals extrapolate less but with longer memory.

Response of beliefs to past returns: positive vs. negative



Schematic: relative decay rates as reported (45% vs. 9% at lag 4); levels illustrative.

Extrapolative Beliefs: Biased Beliefs, Predictability, and Trading Profits

Beliefs negatively predict returns.

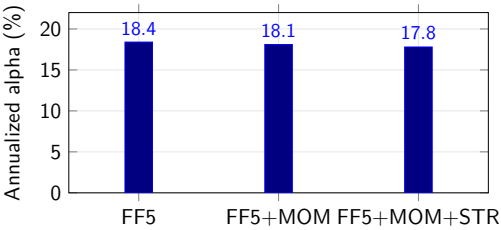
- Higher consensus score ⇒ lower next-week return (Fama–MacBeth); holds for the extrapolation-predicted *and* residual components — beyond short-term reversal.
- Stronger among low-institutional-ownership and high-$\lambda_1(1-\lambda_2)$ stocks.

Model. Extrapolators trade on beliefs; risk-averse fundamental traders only partially correct mispricing ⇒ belief-driven overpricing reverses.

Strategy: long low-score, short high-score.

- ∼7 bps/day ≈ 18%/yr, net of FF5 + momentum + short-term reversal.
- Out-of-sample: predicted scores beat 1-week and 1-month reversal sorts, even among the largest, most liquid non-Forcerank stocks.

Risk-adjusted strategy performance



Reconstructed: ∼18%/yr profit reported as robust across factor specifications; bars show stability, not separate estimates.

Takeaways

1. First direct cross-sectional evidence of *return* (not cash-flow) extrapolation for individual stocks.
2. Extrapolative beliefs are a distinct driver of short-term reversals, beyond liquidity shocks.
3. Limitation: one-week horizon; silent on medium/long-run anomalies without further assumptions.

Over-extrapolation of Past Fundamentals: Application

- Bordalo et al. (2023): the overreaction of long-term earnings growth (LTG) expectations helps explain time-series variation in the P/D ratio and predicts aggregate market returns
- Good news causes investors to become overly optimistic about future earnings growth



- This figure plots LTG (red) and the P/D ratio (green), along with realized stock returns over the coming five years (black)

Bordalo et al. (2023)

Over-extrapolation of Past Fundamentals: Application

- Bordalo et al. (2025): construct a firm-level measure of Expectations Based Returns (EBRs) that uses analyst forecast errors and revisions and shuts down any cross-sectional differences in required returns
- They find variation in EBRs accounts for a large chunk of cross-sectional return spreads in value, investment, size, and momentum factors.

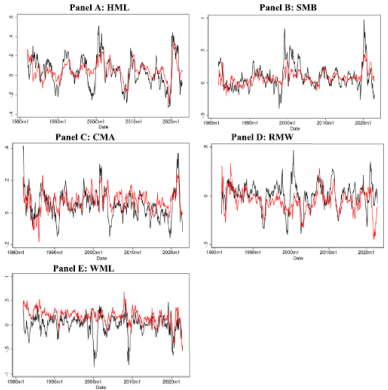


Fig. 1. Actual and expectations-based return spreads.
Note: Panel A plots one-year log returns (black line) and expectations-based returns (EBRs, red line) for the long-short value minus growth (HML) portfolio, where EBRs are computed following Eq. (13) in the text. Panel B plots one-year log returns (black line) and EBRs (red line) for the long-short small minus big (SMB) portfolio. Panel C plots one-year log returns (black line) and EBRs (red line) for the long-short conservative minus aggressive investment (CMA) portfolio. Panel D plots one-year log returns (black line) and EBRs (red line) for the long-short robust minus weak profitability (RMW) portfolio. Panel E plots one-year log returns (black line) and EBRs (red line) for the long-short winners minus losers (WML) portfolio. The sample period is December 1981 to December 2023. See Table 1 for a description of the portfolios.

Bordalo et al. (2025)

My Research: Naïve Earnings Growth Extrapolation

- **Li, Cui, and Zhang (R&R, Review of Accounting Studies)**
  - China reports quarterly earnings on a year-to-date basis, which makes cumulative YTD growth highly salient, even though this YTD measure contains stale earnings information
  - Stocks with the strongest past YTD earnings growth run up in the five days before announcements and reverse afterward
  - Investors extrapolate the salient YTD growth into the upcoming earnings announcement
  - The pattern is stronger for small, retail-heavy, low-coverage stocks, and retail buying demand rises before announcements
- **Connection:** the cross-sectional counterpart of Bordalo et al. (2023) on the previous slide: fundamentals extrapolation that distorts individual stock prices around earnings news, not just the aggregate market

Psychology of Beliefs: Conservatism

- Investors with extrapolative or overconfident beliefs adjust their views too much in response to new information and cause excessive movements in prices
- But some phenomena — momentum and post-earnings announcement drift (PEAD) — may instead require that investors adjust their views *too little*
- **Conservatism**: individuals are too slow to update their beliefs in response to new information (Edwards, 1968)

Psychology of Beliefs: Belief Perseverance and Confirmation Bias

- **Belief perseverance:**
  - Once we have formed an opinion, we are often too slow to change it on receipt of new evidence
  - We do not look for evidence that would falsify our beliefs
  - We ignore evidence that goes against our beliefs
- **Confirmation bias:**
  - We actively seek out evidence that supports our beliefs
  - We interpret ambiguous evidence as supportive of our beliefs

Psychology of Beliefs: Anchoring

- Kahneman and Tversky (1974): when people form numerical estimates of uncertain quantities, their adjustments away from an initial value (called the "anchor") are often insufficient
  - E.g. estimates of the percentage of countries in the United Nations that are African countries were affected by a number between 0 and 100 that was determined by spinning a wheel of fortune in the subjects' presence
  - Those who were asked to compare their estimate to 10 subsequently estimated 25 percent, while those who compared it to 60 estimated 45 percent
- Applications in Finance:
- Anchoring on the 52-week high in the stock market
- George and Hwang (2004): nearness to the 52-week high predicts future returns
- They argue this is because traders are slow to react to good news when the price is close to the 52-week high
- This also explains the momentum effect

The Effect of Feelings

- A recent strand of research investigates how feelings affect beliefs and preferences
  - Affective feelings (mood, emotion)
  - Cognitive feelings
- **Affective feelings**
  - A happier mood caused by an exogenous stimulus leads to more optimistic judgments about unrelated future outcomes (Johnson and Tversky, 1983)
  - E.g. sunshine, sports outcomes, air pollution

Affective Feelings: Applications

- **Sports**
- Edmans, García, and Norli (2007): when the national soccer team loses a World Cup match, the national stock market falls the next day
- Disappointing sports news worsens the national mood, leading people to form more negative judgments about something unrelated, namely the economy's prospects
- **Sunshine/Daylight**
- Hirshleifer and Shumway (2003): the stock market has higher returns on sunnier days
- Kamstra, Kramer, and Levi (2003): fewer hours of daylight lead to a type of depression, Seasonal Affective Disorder (SAD), which, in turn, leads to risk aversion and lower returns

Affective Feelings: Applications, ctd.

- **Terrorist attacks**
- Wang and Yang (2020): Terrorist attacks lead to increased aggregate risk aversion, as proxied by flows into (out of) government bond funds (equity funds)
- These findings point to a more general mechanism that may amplify market fluctuations
  - If the stock market goes up, this improves the mood of many investors, leading them to form more positive judgments about the future prospects of the economy
  - This causes them to push the stock market up still higher

My Research: Air Pollution and Carbon Divestment

- **Li, Huynh, and Xia (RAST 2025)**
  - Mutual fund managers exposed to local air pollution underweight stocks of high-emission firms
  - The effect strengthens for environmentally conscious managers and for those surprised by pollution: managers revise their carbon-risk beliefs in response to an environmental stimulus
  - The divestment is costly: sold carbon-intensive stocks subsequently outperform the stocks bought in their place
- **Connection:** like soccer losses and terrorist attacks, pollution is an exogenous stimulus that shifts beliefs about environmental risks, and here professionals *overreact* by selling carbon-intensive stocks more than is justified by fundamentals

Cognitive Feelings

- The feeling of ease when accessing or processing information leads to more positive judgments than is justified by content alone (Zajonc, 1968; Bornstein, 1989)
- E.g. people like familiar things more than justified by content alone
- In particular, mere exposure to something makes us like it more than is justified
- **Applications:**
  - Home bias in portfolio choice
  - Local bias in stock ownership
  - Excessive allocation to own company stock
- We are exposed, on a daily basis, to our home country, our local region, and the company we work for

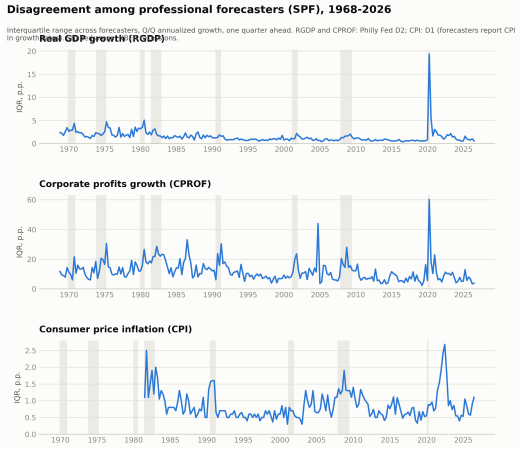
Heterogeneous Beliefs

Heterogeneous Beliefs: Overview

- Previously we looked at the psychological foundations of systematically biased beliefs
  - Assuming a representative investor whose belief is biased
- In this part, we look at models with heterogeneous beliefs
  - Average belief is unbiased
  - But cross-sectional dispersion in beliefs can still affect prices and trading

Heterogeneous Beliefs, ctd.

- People regularly hold different beliefs about almost everything in life
  - from political views and sports events to economic outcomes and asset returns

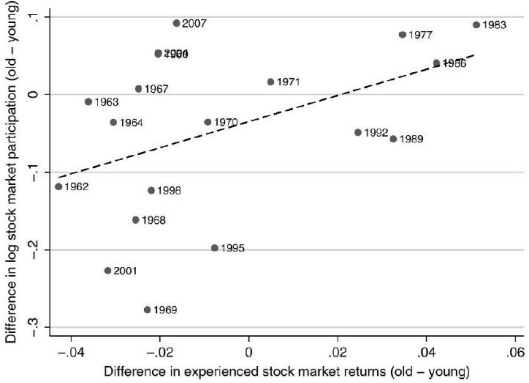


Sources of Heterogeneous Beliefs

- What may cause heterogeneous beliefs?
- Asymmetric information alone cannot cause rational agents with a common prior to agree to disagree
  - e.g., the no-trade theorem of Milgrom and Stokey (1981)
- Heterogeneous beliefs must come from:
  - Differences in prior beliefs: it is reasonable for agents to hold different priors about new things, e.g., Morris (1995)
  - Differences in how agents process information: investors put different weights on the same piece of information
  - The root cause can come from overconfidence, culture, or experience

Sources of Heterogeneous Beliefs — Experience Effects

- One important source of heterogeneous beliefs is personal experience
- When forming beliefs, a rational individual should take account of all available information
- However, mounting evidence suggests that people overweight their personal experience
- Malmendier and Nagel (2011): whether a household participates in the stock market, and how much it invests in it, depends on the stock market returns experienced over its lifetime



Sources of Heterogeneous Beliefs — Experience Effects

- **Inflation expectation** (Malmendier and Nagel, 2016)
- Individuals overweight inflation experienced during their lifetimes when forming expectations about future inflation
- Younger individuals update their beliefs more strongly than older individuals
- Kuchler and Zafar (2019) show that people's beliefs about aggregate economic outcomes strongly depend on their locally experienced economic conditions (housing prices and unemployment)

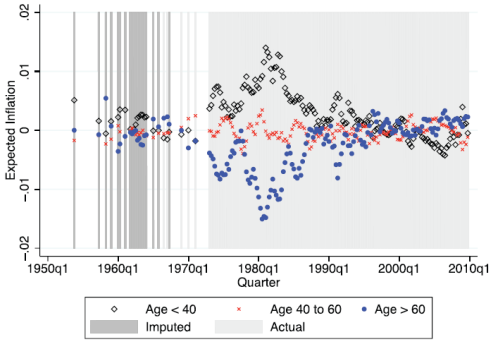


FIGURE I

Inflation Expectations by Age Group Relative to Cross-Sectional Mean

Malmendier and Nagel (2016)

Sources of Heterogeneous Beliefs — Political Polarization

- Goldman, Gupta, and Israelsen (JFE 2025)
- Setting: comparing coverage of the same corporate financial news by the conservative Wall Street Journal and the liberal New York Times
- **Findings:**
  - Newspapers use more positive language and are more likely to report good rather than bad news about politically aligned firms
  - Politically induced disagreement in corporate financial news leads to an increase in abnormal trading volume for the most politically extreme firms

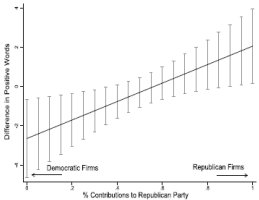


Figure 2A: Difference in Positive Words between WSJ and NYT

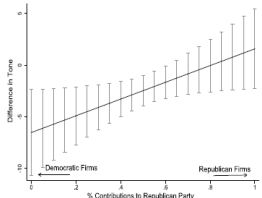


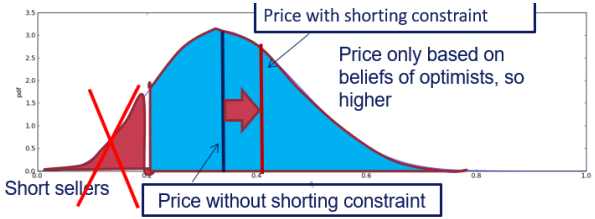
Figure 2B: Difference in Tone between WSJ and NYT

Heterogeneous Beliefs (no SSC)

- **SSC**: short-sale constraints
- Taken alone, overconfidence-based heterogeneous beliefs do not have interesting pricing effects
- The pricing impacts of optimistic and pessimistic investors cancel out
- But they can help explain the high trading volume
  - Investors trade because they disagree about asset values
  - Higher dispersion of beliefs $\Rightarrow$ higher trading volume
- Grinblatt and Keloharju (2009):
  - They use data from Finland to show that more overconfident people trade more
  - Overconfidence is self-reported confidence minus how confident you should be based on performance on aptitude tests

Heterogeneous Beliefs with SSC

- If we combine heterogeneous beliefs with short-sale constraints, we get interesting pricing effects
- **Static argument**
- Miller (1977): heterogeneous beliefs plus a short-sale constraint lead to overpricing

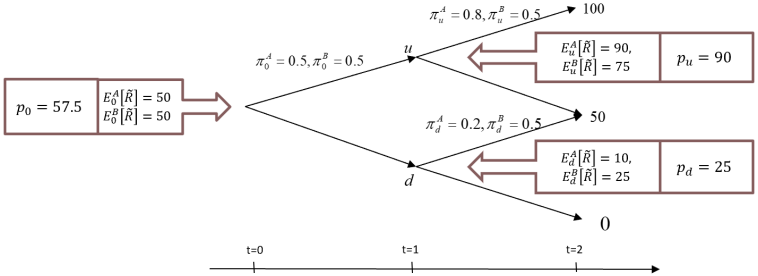


Heterogeneous Beliefs with SSC, ctd.

- **Dynamic argument**
- Harrison and Kreps (1978)
- If there are heterogeneous beliefs, you are willing to pay more than your own valuation of the asset today
  - When information is released, there is a good chance that you will be able to sell to someone more optimistic
  - This creates a "speculative" component to the price
- Scheinkman and Xiong (2003) build on this idea and develop a "resale-option theory" of speculative bubbles
- When short-sales are constrained, an asset owner values the option to resell the asset to other, more optimistic buyers in the future

Heterogeneous Beliefs with SSC, ctd.

- **An example:** $t = 0, 1, 2$
- Risk-neutral investors A and B trade a risky asset
- The interest rate is zero and there is no short selling



Heterogeneous Beliefs with SSC, ctd.

- Both agents' buy-and-hold valuations of the asset are 50
- What is the traded price?
  - In state u, A sets the price at 90
  - In state d, B sets the price at 25
  - Then, at time 0, both value the asset at 57.5!
- This is because both expect to resell the asset to the other agent at a higher price in the future.
  - B expects to resell to A in state u at 90, which is higher than his valuation 75.
  - A expects to resell to B in state d at 25, which is higher than his valuation 10.
- This resale option contributes to both A and B's initial asset valuation: a speculative bubble component.

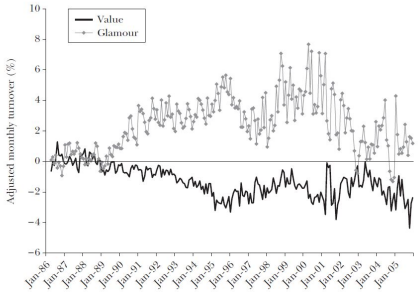
Heterogeneous Beliefs with SSC, ctd.

- Characteristics of this bubble component:
- The valuation of the resale option is determined by fluctuations in agents' beliefs and asset float
- Scheinkman and Xiong (2003): the more likely their beliefs flip, the more they trade and the more valuable the resale option.
- This is why bubbles tend to coincide with trading frenzies.
- Another key factor is asset float: more shares available for trading $\Rightarrow$ smaller bubble
  - This is because when investors have downward-sloping demand curves, an owner can sell his share only if the buyer's belief passes his by a sufficient margin
  - A larger float makes this less likely to happen

Heterogeneous Beliefs with SSC, ctd.

- Models of heterogeneous beliefs with SSC are popular because they not only explain overpricing, but also another important empirical fact:
- **The coincidence of high valuations and heavy trading**
- Hong and Stein (2007): value and growth stocks diverge in both valuation and turnover during the tech bubble period

Figure 2
Turnover in Value and Glamour Stocks, 1986–2005



Heterogeneous Beliefs and Bubbles

- Xiong and Yu (AER 2011) conduct a more rigorous test of the bubble theory based on heterogeneous beliefs — the Chinese warrant bubble
- In 2005–2008, over a dozen put warrants traded in China went so deep out-of-the-money that their intrinsic value was essentially zero
- Yet they traded at high prices with enormous turnover

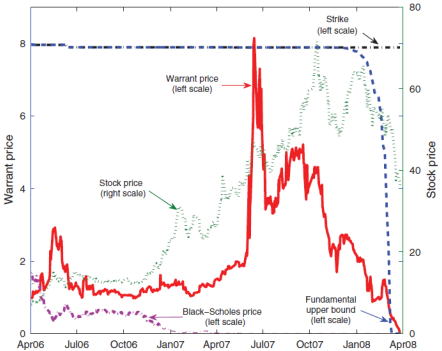


FIGURE 1. PRICES OF WULIANG PUT WARRANT

Notes: This figure shows the daily closing prices of WuLiang stock and its put warrant, along with WuLiang warrant strike price, upper bound of its fundamental value assuming WuLiang stock price drops 10 percent every day before expiration (maximum allowed per day in China's stock market), and its Black-Scholes price using WuLiang stock's previous one-year rolling daily return volatility.

Heterogeneous Beliefs and Bubbles, ctd.

- They find evidence most consistent with the resale option theory of bubbles
- The size of the warrant bubble increases with turnover and volatility, and decreases with asset float

TABLE 5—DETERMINANTS OF SIZE OF THE WARRANTS BUBBLE

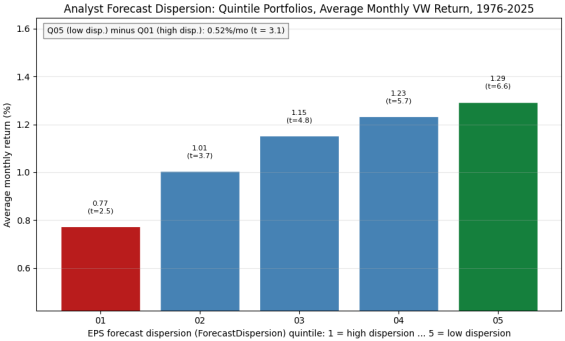
	(1)	(2)	(3)	(4)	(5)	(6)
TURNOVER	0.203			0.158		0.105
(*t*-stat)	(8.91)			(5.46)		(3.80)
VOL		21.76		14.12		13.99
(*t*-stat)		(5.38)		(2.97)		(3.05)
FLOAT			−0.302	−0.324		
(*t*-stat)			(12.62)	(12.31)		
FIRMISSUE					−1.302	−1.260
(*t*-stat)					(25.03)	(23.53)
BROKERAGEISSUE					0.496	0.419
(*t*-stat)					(15.84)	(12.60)

Notes: This table includes the regression results of daily warrant closing prices on *TURNOVER* (daily warrant turnover) in column 1, on *VOL* (5min warrant return volatility) in column 2, on FLOAT (the total number of warrants outstanding, in billions) in column 3, on *TURNOVER*, *VOL*, and *FLOAT* together in column 4, on FIRMISSUE (the total amount of warrants issued by the underlying firm from Table 1, in billions) and *BROKERAGEISSUE* (net total warrant issued by brokerage firms, in billions) in column 5, and on *TURNOVER*, *VOL*, *FIRMISSUE*, and *BROKERAGEISSUE* together in column 6. The regressions use the zero-fundamental sample (defined as Black-Scholes value less than 0.0005, or for cash-settled NanHang if the settlement price will for sure exceed the strike) and include warrant maturity fixed effect. The *t*-statistics, in the parentheses, are adjusted for heteroskedasticity and correlation within a trading day.

Xiong and Yu (AER 2011)

Heterogeneous Beliefs and Cross-Sectional Stock Returns

- The cross-sectional prediction of models of heterogeneous beliefs with short-sale constraints:
- Assets that people disagree about more should be more overpriced
- Diether, Malloy, and Scherbina (JF 2003) test the model's prediction using **analyst forecast dispersion** as a proxy for disagreement
- They find that stocks with higher analyst forecast dispersion earn lower future returns



Machine Forecast Disagreement

Motivation: beliefs are unobservable

- Disagreement drives trading and mispricing: short-sale constraints shut pessimists out ⇒ high-disagreement stocks are overpriced (Miller 1977).
- Classic proxy AFD (analyst dispersion, DMS 2002): limited I/B/E/S coverage; behavioral biases (Johnson 2004).

Construction: MFD

- Each investor k is a random forest: $E_{k,t}[r_{i,t+1}] = RF_k(z_{k,i,t})$.
- K=100 investors share one info set (152 chars.) but each draws a random 50% feature subset (info. differences) + bootstrap/dropout (prior differences) ⇒ spans all three disagreement sources (Barberis 2018).
- Investors are sophisticated yet imperfect; $\mathrm{MFD}_{i,t}$ = dispersion of forecasts across investors.
- 10-yr rolling window, refit annually; full-sample; beats AFD in predictive power.

Results

- Decile long-short (buy low MFD) earns **13%** p.a. value-weighted.
- Earnings-based MFD also predicts; correlates more with AFD.

Mechanism: short-sale constraints, limits to arbitrage

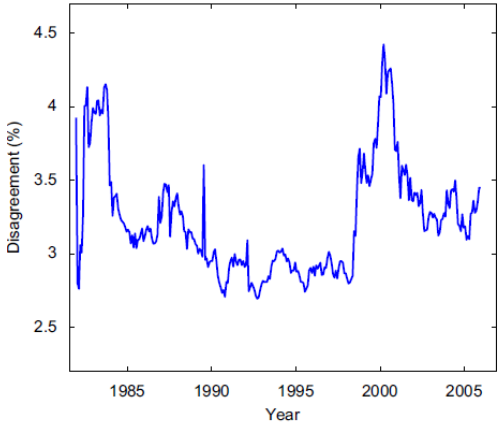
- High borrow-fee -2.36%/mo vs. low-fee -0.12% (diff. -2.24, $t = -6.06$).
- High retail ownership -1.10% vs. inst. -0.31% ($t = 4.53$).
- High-MISP group (Stambaugh–Yu–Yuan): -0.68% vs. -0.31%.
- Correction concentrated on earnings days: long-short 156%/113% higher in 1/3-day windows ⇒ arbitrage limits amplify mispricing.

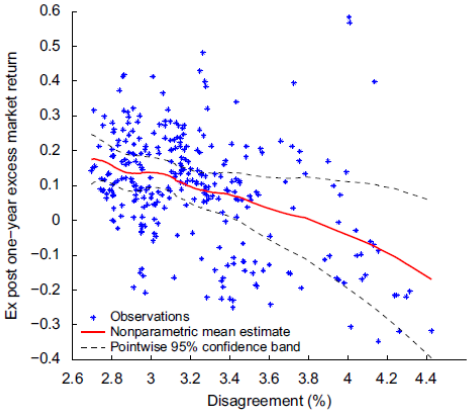
Takeaway: disagreement can be **simulated, not surveyed** — heterogeneous yet rational models generate a belief distribution; high-MFD stocks are overpriced under short-sale constraints ⇒ the discount is mispricing, not risk compensation.

Bali, Kelly, Mörke & Rahman, *Machine Forecast Disagreement*, RFS 2026 (robust to 19 parameterizations).

Heterogeneous Beliefs and the Aggregate Stock Market

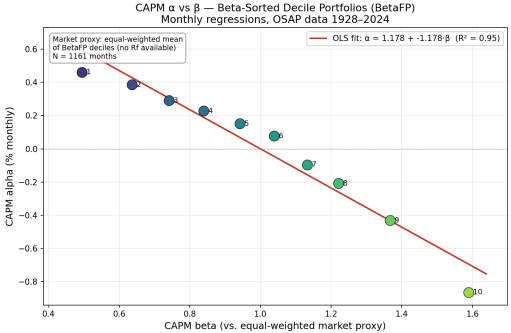
- Yu (2011) tests this prediction for the aggregate stock market
- Aggregate disagreement is measured as the value-weighted average of stock-level dispersion





Heterogeneous Beliefs and Risk-Return Tradeoff

- Another important application: risk-return tradeoff
- Standard theory: higher risk $\Rightarrow$ higher expected return
- But empirically, the relation between risk and return is often negative
- Heterogeneous beliefs models can explain this anomaly



Heterogeneous Beliefs and Risk-Return Tradeoff, ctd.

- Heterogeneous beliefs can also explain the *high beta – low alpha* puzzle
- Hong and Sraer (2016): disagreement about aggregate cash flows plus short-sale constraints

$$\boldsymbol{d}_i = \beta_i(\boldsymbol{Z}) + \boldsymbol{e}_i$$

1. high β assets are more sensitive to disagreement about aggregate cashflows than low β assets
2. Macro-disagreement: optimists $\boldsymbol{Z} + \delta$, pessimists $\boldsymbol{Z} - \delta$
3. high β assets experience more divergence of opinion: optimists $\beta_i(\boldsymbol{Z} + \delta)$ and pessimists $\beta_i(\boldsymbol{Z} - \delta)$
4. then overpricing of high β assets due to short-sale constraints

$\Rightarrow$ CAPM holds when aggregate disagreement is low

$\Rightarrow$ But high disagreement leads to an inverted-U shape: initially increasing with β and then decreasing

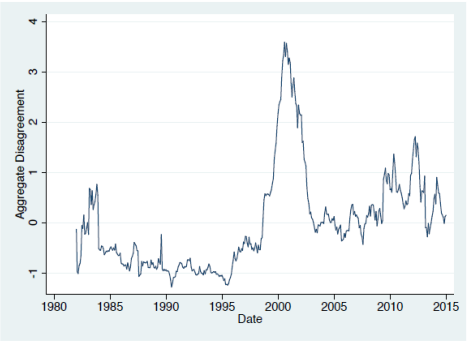
- β ***amplifies aggregate disagreement***

Heterogeneous Beliefs and Risk-Return Tradeoff, ctd.

- Empirical predictions of Hong and Sraer (2016):
- Upward-sloping SML when aggregate disagreement is low
- Inverted-U shape of the SML when disagreement is high
- Stock-level disagreement increases with β, especially when aggregate disagreement is high

Figure 1: Time-series of Aggregate Disagreement

Note: Sample Period: 12/1981-12/2014. Sample: CRSP stock file excluding penny stocks (price < $5) and microcaps (stocks in bottom 2 deciles of the monthly size distribution using NYSE breakpoints). Each month, we calculate for each stock the standard deviation of analyst forecasts on the stock' long run growth of EPS, which is our measure of stock-level disagreement. We also estimate for each stock i $\hat{\beta}_{i,t-1}$, the stock market beta of stock i at the end of the previous month. These betas are estimated with a market model using daily returns over the past calendar year and 5 lags of the market returns. Aggregate Disagreement is the monthly $\hat{\beta}_{i,t-1}$-weighted average of stock-level disagreement.



Heterogeneous Beliefs and Risk-Return Tradeoff, ctd.

Figure: Average 3-months excess returns for equal-weighted beta decile portfolios in low and high aggregate disagreement months.

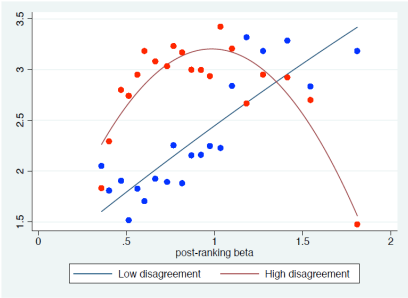
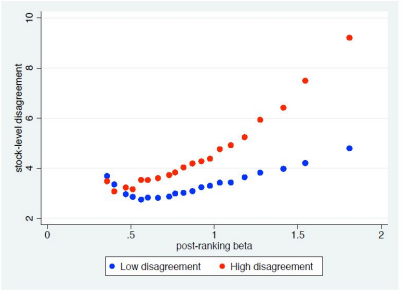
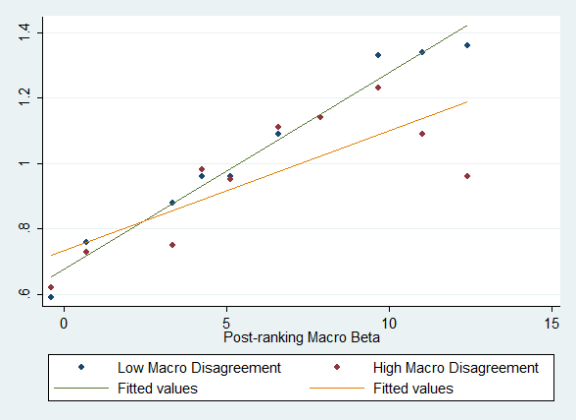


Figure: Equal-weighted average of dispersion of analyst earnings forecasts by beta deciles during low and high aggregate disagreement months.



Heterogeneous Beliefs and Risk-Return Tradeoff, ctd.

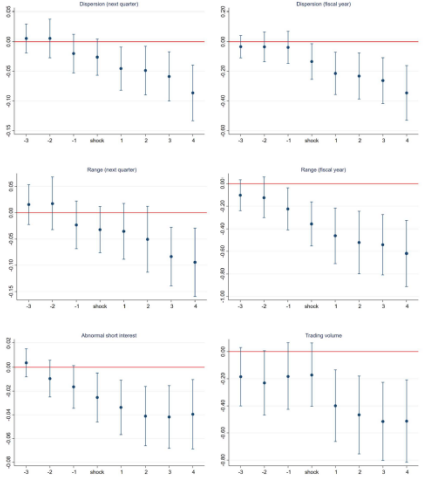
- The same intuition can explain the pricing of macroeconomic factors: my sole-authored work (Li, RAPS 2016)
- High disagreement about a macro factor, measured by forecast dispersion from the Survey of Professional Forecasters, predicts *lower* returns for stocks with high loadings on that factor



The Causal Effect of Disagreement on Asset Prices

- Identifying exogenous shocks to investor disagreement is still challenging
- Chang, Hsiao, and Ljungqvist (JF 2022) exploit the staggered implementation of EDGAR by the SEC, which induces a reduction in information acquisition costs
  - Before EDGAR, investors could access firms' mandatory filings only at high cost, either by subscribing to commercial data providers or by physically visiting the SEC's reference rooms
  - Beginning in April 1993, the SEC required U.S. firms to file their mandatory disclosures electronically through the EDGAR system
  - SEC randomly assigned firms to one of ten implementation waves, thereby staggering inclusion in EDGAR over a three-year period between 1993 and 1996
- EDGAR inclusion helps resolve disagreement around information events, leading to price corrections and lower crash risk

The Causal Effect of Disagreement on Asset Prices, ctd.



	Skewness (*NCSKEW*) (1)	Down-to-up volatility (*DUVOL*) (2)	Extreme negative returns, 0.01% (*CRASH*001) (3)
Quarter of EDGAR inclusion	0.007	0.000	-0.003
	0.022	*0.014*	*0.008*
Next four quarters	-0.049**	-0.031**	-0.027***
	0.020	*0.012*	*0.007*
Controls?	yes	yes	yes
Calendar quarter FE?	yes	yes	yes
Fiscal quarter FE?	yes	yes	yes
Firm FE?	yes	yes	yes
R-squared	17.4%	19.1%	15.8%
Pre-trends (*p*-value)	0.562	0.141	0.957
No. of firms	3,366	3,366	3,366
No. of firm-quarters	28,652	28,652	28,652

Biased Beliefs: Concerns

- Do people simply learn their way out of biased beliefs?
- Only to a limited extent
- Many of the most persistent behavioral biases are mental shortcuts that were useful for survival over human evolution
  - "Loss aversion" increases survival probability
  - Overconfidence increases effort and trust within your community
- Do highly trained professionals exhibit these biases?
  - Yes, professionals sometimes do worse
  - CEOs and founders are more overconfident than the average person
- Do people exhibit these biases even when stakes are high?

Stakes and Investor Biases

- Sui and Wang (JFE 2025): investors do better when they care less
- Unique setting where investors trade stocks in real accounts using their own money and simultaneously in a simulated setting
- Within-investor estimation shows that investors exhibit stronger biases and perform worse in higher-stakes real accounts than in lower-stakes simulated accounts
- Explanation: individuals carry incorrect mental models and greater attention/effort leads to the more careful application of those incorrect models

Table 6

Differences between the real and simulated accounts. In this table, we report differences between real and simulated accounts. See Section 4.1 for the bias definitions. For each type of bias we report the mean of the real accounts, the mean of the simulated accounts, and the mean of the differences between them. For the differences, we report *t*-statistics and Wilcoxon *p*-values. In column "N" we report the number of investors.

		Real	Simulated	Difference		
	N	Mean	Mean	Mean	*t*-stat	Wilcoxon *p*
Disposition effect	4371	4.163	0.489	3.674	(32.43)	0.000
Lottery preference	4369	6.491	4.264	2.228	(13.88)	0.000
Extrapolation	4373	15.599	12.66	2.939	(6.49)	0.006
Underdiversification	4394	−1.086	−1.473	0.386	(32.67)	0.000
Local Bias	1739	0.630	0.645	−0.015	(−0.64)	0.810
Turnover	4306	5.960	0.839	5.121	(47.44)	0.000
Performance	4375	−0.656	−0.148	−0.509	(−4.95)	0.001